

# APOLLO

A Formal Specification of the Market-to-Music Pipeline:  
Spectral Analysis, Generative Sampling, and the Sonification Grammar of *TECH(NO)*  
*US*

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## Abstract

We give a complete mathematical specification of a pipeline that maps a financial price history to a musical composition. Part I establishes the second-order theory: the Wiener–Khinchin representation of the power spectral density (PSD), its estimators (periodogram, Welch, Blackman–Tukey, Thomson multitaper), Fisher’s exact test for hidden periodicities, and the Geweke–Porter–Hudak (GPH) estimator of long memory. Part II specifies three generative models admitting exact or approximate sampling from the estimated law: phase-randomized surrogates, GJR-GARCH with skew- $t$  innovations, and a recurrent conditional-density network. Part III defines the sonification grammar: a measurable map from the path of the (real or sampled) return process to a discrete musical event set and thence to a pressure waveform, under which distinct moments of the conditional distribution are carried by perceptually orthogonal audio dimensions — location by pitch, scale by loudness, and tail mass by timbre.

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\*Pipeline implementation: [github.com/vitociciretti/apollo](https://github.com/vitociciretti/apollo). This document specifies the mathematics implemented in `pipeline/`.

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## 1 Data and notation

Let  $P_t > 0$ ,  $t = 0, 1, \dots, n$ , denote the dividend- and split-adjusted daily closing price of an asset on trading-day grid  $t$ . Define log-returns

$$r_t := \log P_t - \log P_{t-1}, \quad t = 1, \dots, n. \quad (1)$$

Throughout,  $\hat{\mu} := n^{-1} \sum_t r_t$ ,  $\hat{s}^2 := n^{-1} \sum_t (r_t - \hat{\mu})^2$ , and the *volatility proxy* is  $a_t := |r_t|$ . Annualization uses the convention  $\sigma_{\text{ann}} = \hat{s} \sqrt{252}$ . The cumulative log-price is  $X_t := \sum_{u \leq t} r_u$ ; the running maximum  $M_t := \max_{u \leq t} X_u$ ; and the *drawdown process*

$$D_t := M_t - X_t \geq 0. \quad (2)$$

**Assumption 1.1.**  $(r_t)$  is covariance-stationary with autocovariance  $\gamma(h) = \text{Cov}(r_t, r_{t+h})$  and  $\sum_h |\gamma(h)|$  possibly divergent (long memory is admitted for  $a_t$ ; see §2.3).

## 2 Second-order structure and the Wiener–Khinchin theorem

**Theorem 2.1** (Wiener–Khinchin). *Let  $(Y_t)_{t \in \mathbb{Z}}$  be covariance-stationary with autocovariance  $\gamma(\cdot)$ . There exists a finite measure  $F$  on  $[-1/2, 1/2]$  (the spectral measure) with*

$$\gamma(h) = \int_{-1/2}^{1/2} e^{2\pi i f h} dF(f), \quad h \in \mathbb{Z}. \quad (3)$$

If  $\sum_h |\gamma(h)| < \infty$ ,  $F$  is absolutely continuous with density

$$S(f) = \sum_{h=-\infty}^{\infty} \gamma(h) e^{-2\pi i f h}, \quad f \in [-1/2, 1/2], \quad (4)$$

the power spectral density, and  $\gamma(h) = \int e^{2\pi i f h} S(f) df$ ; in particular  $\text{Var}(Y_t) = \int_{-1/2}^{1/2} S(f) df$ .

Equation (4) is the load-bearing identity of the project: the PSD is the Fourier transform of the autocovariance, hence estimating  $S$  and estimating  $\gamma$  are the same problem, and any generative model matching  $S$  matches all second-order properties of the process.

### 2.1 Estimators implemented

Write  $Y_t$  for the demeaned series of interest ( $r_t - \hat{\mu}$  or  $a_t - \bar{a}$ ).

**(i) Periodogram.** With the DFT  $d(f_j) = \sum_{t=1}^n Y_t e^{-2\pi i f_j t}$  at Fourier frequencies  $f_j = j/n$ ,  $j = 1, \dots, m = \lfloor (n-1)/2 \rfloor$ ,

$$I(f_j) = \frac{1}{n} |d(f_j)|^2. \quad (5)$$

$I$  is asymptotically unbiased for  $S$  but inconsistent ( $I(f_j)/S(f_j) \xrightarrow{d} \frac{1}{2}\chi_2^2$ , variance not vanishing), so it is used only where its *exact distribution* is exploited (Fisher's test, §2.2; GPH, §2.3).

**(ii) Welch.** Split  $Y$  into  $K$  segments of length  $L$  (implementation:  $L = 512$ , 75% overlap), taper each with a Hann window  $w$ , and average the windowed periodograms:

$$\hat{S}_W(f) = \frac{1}{K} \sum_{k=1}^K \frac{1}{\sum_t w_t^2} \left| \sum_{t=1}^L w_t Y_{t+o_k} e^{-2\pi i f t} \right|^2. \quad (6)$$

Variance decays like  $1/K$  at the cost of frequency resolution  $\asymp 1/L$ .

**(iii) Blackman–Tukey (the Wiener–Khinchin route).** Estimate  $\hat{\gamma}(h) = n^{-1} \sum_{t=1}^{n-h} Y_t Y_{t+h}$  for  $|h| \leq L$  (implementation:  $L = 512$ ), damp with the Tukey–Hanning lag window  $\lambda(h) = \frac{1}{2}(1 + \cos \frac{\pi h}{L})$ , and transform:

$$\hat{S}_{BT}(f) = \sum_{|h| \leq L} \lambda(h) \hat{\gamma}(h) e^{-2\pi i f h}. \quad (7)$$

This estimator is Theorem 2.1 made operational; agreement of  $\hat{S}_{BT}$  with  $\hat{S}_W$  is used as an internal consistency check.

**(iv) Thomson multitaper.** Let  $v^{(0)}, \dots, v^{(K-1)}$  be the discrete prolate spheroidal sequences (DPSS) of length  $n$  and time–bandwidth product  $NW$  (implementation:  $NW = 4$ ,  $K = 7$ ), the orthonormal maximizers of spectral concentration in  $[-W, W]$ . The estimator

$$\hat{S}_{MT}(f) = \frac{1}{K} \sum_{k=0}^{K-1} \left| \sum_{t=1}^n v_t^{(k)} Y_t e^{-2\pi i f t} \right|^2 \quad (8)$$

has  $\approx 2K$  equivalent degrees of freedom and minimal broadband leakage, making it the estimator of record for detecting weak spectral *lines* against a colored background — precisely the regime of volatility spectra.

## 2.2 Fisher's exact test for hidden periodicities

Under  $H_0$ :  $r_t \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$ , the normalized ordinates  $I(f_j)/\sum_k I(f_k)$  are exchangeable and Fisher's statistic

$$g = \frac{\max_{1 \leq j \leq m} I(f_j)}{\sum_{j=1}^m I(f_j)} \quad (9)$$

has the exact null law

$$\Pr(g > x) = \sum_{k=1}^{\lfloor 1/x \rfloor} (-1)^{k-1} \binom{m}{k} (1 - kx)^{m-1}. \quad (10)$$

A per-ordinate 5% false-alarm level (Bonferroni across  $m$  ordinates) is  $x^* = 1 - (0.05/m)^{1/(m-1)}$ .

**Empirical result (NVDA, 2010–2026,  $n = 4156$ ).** For  $r_t$ :  $g$  attains  $p = 0.34$  and zero ordinates clear  $x^*$  — the return spectrum is statistically flat (white noise), consistent with the martingale hypothesis. For  $a_t = |r_t|$ :  $p < 10^{-4}$  with eight ordinates above the false-alarm level, all at low frequency, including periods  $\approx 63$  and  $\approx 126$  trading days — the quarterly and semi-annual earnings-calendar harmonics.

**Remark 2.2.** *Once long memory in  $a_t$  is established (§2.3) the white-noise null is weak for individual low-frequency peaks: power concentration under a  $1/f$ -shaped spectrum inflates (10) rejections. The robust finding is the long-memory slope plus the calendar harmonics, not any single line.*

### 2.3 Long memory: fractional integration and GPH

**Definition 2.3.**  $(Y_t)$  has long memory with parameter  $d \in (0, \frac{1}{2})$  if  $S(f) \sim c f^{-2d}$  as  $f \downarrow 0$ , equivalently  $\gamma(h) \sim c' h^{2d-1}$  as  $h \rightarrow \infty$ .

The Geweke–Porter–Hudak estimator regresses log-periodogram ordinates on the canonical regressor. With  $\lambda_j = 2\pi f_j$  and bandwidth  $m^* = \lfloor n^{0.6} \rfloor$ ,

$$\log I(f_j) = c - d \log(4 \sin^2(\lambda_j/2)) + \varepsilon_j, \quad j = 1, \dots, m^*, \quad (11)$$

estimated by OLS; asymptotically  $\hat{d} \approx \mathcal{N}(d, \pi^2/(24 m^*))$ , i.e.  $\text{se}(\hat{d}) = \pi/\sqrt{24 m^*}$ .

**Empirical result.** For NVDA  $a_t$ :  $\hat{d} = 0.397$ ,  $\text{se} = 0.053$  ( $t = 7.5$ ), squarely in the FIGARCH range documented for equity volatility. For  $r_t$  the low-frequency log–log slope is  $-0.04$  (white noise = 0).

## 3 Generative models: sampling from the estimated law

The composition engine (§4) consumes an arbitrary path  $(r_t^*)_{t \leq N}$ ; three samplers of increasing structural fidelity are implemented.

### 3.1 Phase randomization (maximum entropy given the ACF)

Let  $A_j = |\sum_t (a_t - \bar{a}) e^{-2\pi i j t/n}|$  be the empirical amplitude spectrum, interpolated to target length  $N$  and rescaled by  $\sqrt{N/n}$ . Draw  $\varphi_j \stackrel{iid}{\sim} U[0, 2\pi)$ ,  $\varphi_0 = 0$ , and set

$$a_t^* = \bar{a} + \frac{1}{N} \sum_j A_j e^{i\varphi_j} e^{2\pi i j t/N}, \quad a_t^* \leftarrow \max(a_t^*, 0). \quad (12)$$

By construction  $(a_t^*)$  has (up to interpolation error) the sample autocovariance of  $(a_t)$ ; among all processes with that autocovariance the Gaussian one generated here maximizes entropy. This is Theorem 2.1 used as a *sampler*: the PSD is a complete generative model for the second-order law.

### 3.2 GJR-GARCH(1, 1, 1) with skew- $t$ innovations

$$r_t = \mu + \sigma_t z_t, \quad z_t \stackrel{iid}{\sim} \text{skew-}t(\eta, \lambda), \quad (13)$$

$$\sigma_t^2 = \omega + (\alpha + \gamma \mathbf{1}\{r_{t-1} < \mu\}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \varepsilon_t = r_t - \mu, \quad (14)$$

estimated by maximum likelihood. The leverage term  $\gamma > 0$  generates the volatility asymmetry that the linear samplers of §3.1 cannot. Fitted on NVDA ( $\times 100$  scale):  $\hat{\mu} = 0.116$ ,  $\hat{\omega} = 0.073$ ,  $\hat{\alpha} = 0.030$ ,  $\hat{\gamma} = 0.070$ ,  $\hat{\beta} = 0.932$ ,  $\hat{\eta} = 4.57$ ,  $\hat{\lambda} = -0.02$ ; persistence  $\hat{\alpha} + \hat{\gamma}/2 + \hat{\beta} \approx 0.997$ .

### 3.3 Recurrent conditional-density model

Let  $x_t = (r_t/\hat{s}, \log(|r_t|/\hat{s} + 10^{-3})) \in \mathbb{R}^2$ . A single-layer LSTM  $h_t = \text{LSTM}(x_t, h_{t-1}) \in \mathbb{R}^{64}$  feeds an affine head producing the parameters of the *conditional law of tomorrow's standardized return*:

$$\mu_t = 0.05 \tanh(w_\mu^\top h_t), \quad \sigma_t = \exp(\text{clip}(w_\sigma^\top h_t, [-4, 2])), \quad \nu = 2.05 + \text{softplus}(\theta_\nu), \quad (15)$$

$$\frac{r_{t+1}}{\hat{s}} \Big| \mathcal{F}_t \sim \mu_t + \sigma_t T_\nu, \quad (16)$$

with  $T_\nu$  standard Student- $t$ . Training minimizes the exact negative log-likelihood

$$-\ell = -\log \Gamma\left(\frac{\nu+1}{2}\right) + \log \Gamma\left(\frac{\nu}{2}\right) + \frac{1}{2} \log(\pi\nu) + \log \sigma_t + \frac{\nu+1}{2} \log\left(1 + \frac{z_t^2}{\nu}\right), \quad z_t = \frac{r_{t+1}/\hat{s} - \mu_t}{\sigma_t}, \quad (17)$$

over overlapping windows (length 256, stride 16, 32-step burn-in excluded from the loss), Adam with gradient clipping, model selection by validation NLL on a 15% holdout. Sampling is autoregressive: the hidden state is warmed on the final 252 real observations, then  $r_{t+1}^* = \hat{s}(\mu_t + \sigma_t T_\nu)$  is drawn, clipped at 8 standard units, and fed back through  $x_{t+1}$ .

**Diagnostics (real vs. sampled,  $N = 4608$ ).**

|                        | NVDA             | LSTM sample    |
|------------------------|------------------|----------------|
| tail d.o.f.            | 4.57 (GARCH MLE) | 4.24 (learned) |
| excess kurtosis basis  | 9.8              | 10.1           |
| ACF $_{ r }(1)$        | 0.171            | 0.131          |
| ACF $_{ r }(21)$       | 0.092            | 0.089          |
| GPH $\hat{d}$ on $ r $ | 0.397            | 0.516          |
| annualized vol         | 45.2%            | 53.4%          |
| mean daily return      | +0.148%          | +0.044%        |

Fat tails, volatility clustering and long memory are recovered from raw likelihood training; the conditional mean is (correctly) shrunk toward zero, which materially darkens the sampled paths' drawdown structure.

## 4 The sonification grammar

### 4.1 Grids and state processes

Fix beats per minute  $B$  and the sixteenth-note duration  $\tau := \frac{60}{4B}$  seconds. Days are binned  $\delta$  per sixteenth,  $\delta = \max(2, \text{round}(n/1536))$ , giving  $K = 16 \lfloor n/(16\delta) \rfloor$  sixteenth-notes and  $K/16$  bars. Define on the day grid: rolling volatility  $v_t = \text{sd}(r_{t-20:t})\sqrt{252}$ ; rolling kurtosis  $\kappa_t = \widehat{\text{kurt}}(r_{t-62:t})$ ; trend z-score

$$z_t = \frac{X_t - \text{MA}_{252}(X)_t}{\text{sd}_{252}(X)_t}, \quad (18)$$

and the *regime indicator* from the drawdown (2),

$$\beta_t = \mathbf{1}\{D_t > Q_{0.78}(D)\}, \quad (19)$$

i.e. breakdown  $\equiv$  worst-quartile drawdown *of the path's own distribution* (adaptive by construction: any sampled world keeps  $\approx 22\%$  breakdown time). Bar-level states  $(z_b, v_b, \kappa_b, \beta_b)$  evaluate these at each bar's final day;  $v_b$  is min-max normalized to  $\tilde{v}_b \in [0, 1]$  and  $\kappa_b$  is mapped to its percentile rank  $\tilde{\kappa}_b \in [0, 1]$ .

## 4.2 Pitch: scale quantization

Let the key root be pitch class  $\rho$  (per track, Table 1) and octave-anchored MIDI root  $R$ . Modes are  $\mathcal{S}_{\text{aeol}} = \{0, 2, 3, 5, 7, 8, 10\}$  and  $\mathcal{S}_{\text{phry}} = \{0, 1, 3, 5, 7, 8, 10\}$ , selected by  $\beta$ :  $\mathcal{S}_b = \mathcal{S}_{\text{phry}}$  if  $\beta_b = 1$  else  $\mathcal{S}_{\text{aeol}}$ . The harmony degree per bar is

$$\deg_b = \begin{cases} 5 & z_b < -1, \\ 0 & -1 \leq z_b \leq 0.5, \\ 3 & z_b > 0.5, \end{cases} \quad m_b = R + \mathcal{S}_b[\deg_b]. \quad (20)$$

Pitch is thus a quantized functional of price-versus-trend; the mode (and hence the semitone content) is a functional of the drawdown regime.

## 4.3 Event maps

For sixteenth  $k$  in bar  $b$  with day window  $W_k = [k\delta, (k+1)\delta)$ :

$$\begin{aligned} \textbf{kick:} \quad & k \bmod 4 = 0 \wedge \beta_b = 0 && \text{velocity} \propto 0.85 + 0.15 \tilde{v}_b, \\ \textbf{bass note:} \quad & \beta_b = 0, \text{ pitch } m_b + 12 \cdot \mathbf{1}_{\{\max_{t \in W_k} |r_t| > \hat{s} \wedge k \bmod 4 = 2\}}, && \text{off-beat accent} \propto 1.15, \\ \textbf{stab (chord):} \quad & k \bmod 16 \in \{3, 7, 10, 14\} \wedge \sum_{t \in W_k} |r_t| > \delta \overline{|r|}, && \text{triad on } \deg_b, \\ \textbf{clap roll:} \quad & \beta_b = 1 \wedge \beta_{b+1} = 0 && \text{velocity ramp in } k \bmod 16, \\ \textbf{crash:} \quad & \max_{t \in W_k} |r_t| > 4\hat{s} \vee (\text{first } k \text{ of a recovery bar}). \end{aligned}$$

The filter-cutoff automation (MIDI CC74; audio-domain lowpass corner) is

$$c_b = 160 + 2800 \tilde{v}_b + 500 \tilde{\kappa}_b \text{ Hz}, \quad (21)$$

so the audible brightness of the bass *is* the volatility curve.

## 4.4 Timbre as a tail functional

Each melodic voice is a detuned additive synth

$$y(t) = \sum_{\pm} \sum_{h=1}^H \frac{\alpha_h}{h} \sin(2\pi h f(1 \pm \epsilon)t), \quad \alpha_h = \tilde{\kappa} \mathbf{1}_{\{h \geq 2\}}, \quad \epsilon = 0.004 + 0.006 \tilde{\kappa}, \quad (22)$$

band-limited at the filter corner: at  $\tilde{\kappa} = 0$  a pure sine (thin-tailed, quasi-Gaussian regime), at  $\tilde{\kappa} = 1$  a bright sawtooth (crash-prone regime). Thus the fourth moment is carried by harmonic content — the “piano vs. violin” axis — while the second moment is carried by loudness and the first by pitch: three perceptually orthogonal channels for three moments.

## 4.5 Pads: the spectrum made audible

For each significant volatility-spectrum peak at frequency  $f_0$  (period  $1/f_0 \in [40, 600]$  days), the band-limited analytic envelope is extracted by zero-phase Butterworth bandpass  $\mathcal{B}_{[f_0/1.35, 1.35f_0]}$  and the Hilbert transform  $\mathcal{H}$ :

$$E_{f_0}(t) = \text{smooth} \left| (\text{id} + i\mathcal{H}) \mathcal{B}_{[f_0/1.35, 1.35f_0]}(a - \bar{a})(t) \right|, \quad (23)$$

i.e. the instantaneous amplitude of that volatility cycle through time. The pad voice sounds at the peak frequency transposed into the audible range,  $f_{\text{aud}} = 2.2 \times 10^5 f_0$  Hz, quantized to the track’s scale lattice  $\{440 \cdot 2^{(m-69)/12} : m \bmod 12 \in \rho + \mathcal{S}\}$ , amplitude-modulated by  $E_{f_0}$ , and gated by the smoothed regime indicator (19) so the spectrum is heard precisely when the beat is absent.

## 4.6 Mixing operators

Sidechain compression is the multiplicative envelope  $g(t) = 1 - 0.55 e^{-(t-t_{\text{kick}})/0.085}$  applied to all non-drum buses after each kick onset; spatialization applies a dotted-eighth ( $3\tau$ ) cross-feedback delay to the  $>1.2$  kHz band; mastering is a 25 Hz DC-blocking highpass and peak normalization (no nonlinear stage — an earlier tanh soft-clipper was removed as it injected broadband intermodulation audible as background noise).

## 5 Album parameterization

Tempo is an affine function of annualized volatility,  $B = \text{clip}(\lfloor 120 + 40(\sigma_{\text{ann}} - 0.20) \rfloor, 122, 134)$ ; each asset receives a fixed key.

| #  | Title            | Key    | $\sigma_{\text{ann}}$ | BPM |
|----|------------------|--------|-----------------------|-----|
| 01 | Chipmaker        | F min  | 45%                   | 130 |
| 02 | Cupertino        | A min  | 28%                   | 123 |
| 03 | Redmond          | D min  | 26%                   | 122 |
| 04 | PageRank         | G min  | 28%                   | 123 |
| 05 | Everything Store | C min  | 33%                   | 125 |
| 06 | Social Graph     | B♭ min | 40%                   | 127 |
| 07 | Stream           | E min  | 49%                   | 131 |
| 08 | Ludicrous        | B min  | 57%                   | 134 |

Table 1: *TECH(NO) US*: per-track parameters derived from each asset’s history.

## 6 v4: form, melody, dynamics, and the choir

Version 4 of the engine adds four data-driven layers on top of the grammar of §4 (implementation: `techno_album.py` with the modules `regime.py`, `motif.py`, `volume_perc.py`, `choir.py`): musical *form* from a regime segmentation, a per-asset *theme* with development operators, a trading-*volume* percussion channel, and a sung *choir*. Two tracks extend the album beyond Table 1: 09 (*Correlation One*, the two-asset duet in F minor) and 10 (*Descensus Gradientis*, F minor, 145 BPM), composed not from a price path but from the 80-epoch training history of the recurrent conditional-density model of §3.3.

### 6.1 Regime segmentation: the change-point rule

Let  $n_B = K/16$  be the number of bars and evaluate, at each bar’s final day, the annualized 21-day rolling volatility  $v_b$  and the regime indicator  $\beta_b$  of (19). Define the bar-level volatility percentile rank  $\pi_b = \text{rank}(v_b)/(n_B - 1) \in [0, 1]$ . A new *era* opens at bar  $b$  when the percentile has jumped

$$|\pi_b - \pi_{b-w}| > c, \quad w = 4 \text{ bars}, \quad c = 0.25, \quad (24)$$

subject to a minimum era length of 6 bars (candidate boundaries closer than 6 bars to a previous boundary or to either edge are dropped). Eras are classified **calm/build/peak** by the tercile of their mean  $\pi$  over non-bear bars; maximal runs of  $\beta_b = 1$  of length  $\geq 2$  become **bear** sections (single-bar excursions are treated as noise, and single-bar gaps inside a bear run are bridged), with the final 1–2 bars of each terminating run relabelled **recovery**; the first 4 and last 2 bars are forced **intro/outro**, and non-structural slivers shorter than 5 bars are absorbed into their neighbours. Each section kind carries a *mask* — Boolean gates per layer

(kick/bass/stabs/lead/pads), hat/percussion density multipliers, a half-time flag — and an energy  $e \in \{0.15, \dots, 1.0\}$ ; the music bus is gain-staged *before* the effects chain by

$$g_b = 0.55 + 0.45 \operatorname{clip}\left(\frac{e_b - 0.15}{0.85}, 0, 1\right), \quad (25)$$

linearly interpolated across bar centres, so quiet sections are genuinely quiet. Section downbeats receive a crash; a one-bar sixteenth-note roll fills into every peak section.

## 6.2 The theme and its development operators

**Derivation.** Each asset owns one theme  $\theta \in \{0, \dots, 13\}^8$  (two diatonic octaves of scale degrees), derived deterministically from its *first trading year*: the cumulative log-price  $X_{1:252}$  is reduced to 8 segment means, min–max normalized to  $[0, 1]$ , rounded to degrees 0..9, then musicality is enforced —  $\theta_1 = 0$  (tonic anchor), successive leaps clamped to  $\leq 4$  degrees, and the cadence pulled until  $|\theta_8 - \theta_1| \leq 2$ . The eight resulting themes are distinct: the melody of each track is literally the shape of that stock’s first year.

**Development.** Per bar  $b$ , one of five maps  $\{0, \dots, 13\}^8 \rightarrow \{0, \dots, 13\}^8$  is selected by the bar state  $(\beta_b, z_b, \pi_b)$ , first match wins:

- inversion** ( $\beta_b = 1$ ):  $\mathcal{I}(\theta)_i = (\min_j \theta_j + \max_j \theta_j) - \theta_i$ ,
- transposition** ( $|z_b| > 1.5$ ):  $\mathcal{T}(\theta)_i = \operatorname{clip}(\theta_i + \lfloor z_b \rfloor, 0, 13)$ ,
- diminution** ( $\pi_b > 0.8$ ):  $\mathcal{D}(\theta) = \theta$  on a half-bar grid, played twice,
- augmentation** ( $\pi_b < 0.3$ ):  $\mathcal{A}(\theta) = \theta_{1:4}$  or  $\theta_{5:8}$  (bar parity), doubled durations,
- prime** (else):  $\operatorname{id}(\theta)$ .

The inversion is an exact mirror in the theme’s own compass; a stretched trend shifts the register; high volatility doubles time; calm stretches it. In the audio engine the bear-bar inversion is realized *chromatically*: degrees are first mapped to MIDI via the home scale and mirrored around the fixed integer pivot  $m_{\min} + m_{\max}$  of the theme’s own compass, so every semitone interval is exactly negated (degree-space mirrors do not negate intervals, scale steps being uneven). In bear sections the lead is the featured voice: the inverted theme over the pads.

## 6.3 Trading volume as percussion density

Let  $V_t$  be daily log-volume, aligned 1:1 with  $r_t$ . The per-bar density is the smoothed percentile rank of bar-mean log-volume,

$$d_b = \operatorname{MA}_3\left(\frac{\operatorname{rank}(\bar{V}_b)}{n_B - 1}\right) \in [0, 1], \quad \bar{V}_b = \text{mean of } V_t \text{ over bar } b. \quad (26)$$

Hat velocities scale by the factor  $0.55 + 0.60 d_b$ , and the odd-sixteenth percussion ladder opens position by position as  $d_b$  exceeds fixed thresholds in  $\{0.25, \dots, 0.90\}$ : heavy tape is audible as busy percussion. Separately, with the local spike  $s_t = V_t - \operatorname{MA}_{21}(V)_t$ , a bar is an *earnings-fill* bar when its maximum single-day spike lies in the top decile of per-bar maxima ( $\approx n_B/10$  fills per track); such bars get a snare/clap fill in the last beat, suppressed where the kit is gated off or a section fill already fires.

## 6.4 The key-modulation trigger

Define the bull indicator at bar  $b$  as  $\text{bull}_b = \mathbf{1}\{z_b > 1\}$  with  $z_b$  the 252-day trend z-score of §4 evaluated at the bar’s final day. A section  $[b_0, b_1)$  that is not **bear** and satisfies  $\text{bull}_b = 1$  for *every* bar modulates melody and stabs to the *relative major*: root  $\rho + 3$ , scale  $\mathcal{S}_{\text{maj}} = \{0, 2, 4, 5, 7, 9, 11\}$ . The lift is section-quantized by construction — a strong bull regime is heard as the whole passage brightening, and the home minor returns when the regime ends.

## 6.5 The choir: TTS → phase vocoder → stacked vocoder

The sung voice of tracks 01/07/09/10 is a fully synthetic pipeline (`choir.py`); no human singer is recorded.

1. **TTS.** Each lyric phrase is rendered by a local text-to-speech voice (a locally installed `piper` model when available, else Windows SAPI), giving a spoken take  $x(t)$ .
2. **Segmentation.** The take is split into as many syllable-like chunks as there are melody notes: silence-trimmed, then boundaries placed at energy minima of the smoothed envelope near the equal-split positions; chunks with no voiced core extend backward over the previous vowel (melisma rule).
3. **Pitch mapping.** Per chunk, the fundamental  $f_0$  is measured by windowed autocorrelation (with octave disambiguation and parabolic refinement). The chunk is phase-vocoder time-stretched — STFT with hop-ratio magnitude interpolation and phase advance accumulation — to its note duration, and resampled by  $2^{s/12}$  with  $s = \text{clip}(m - \text{midi}(f_0), \pm 8 \text{ st})$ , where  $m$  is the melody note; the clamp protects formants, and when the melody median sits more than 5 st from the speaking register the TTS source is first re-rendered at a shifted SSML prosody pitch so the clamp never binds. Voiced sub-blocks absorb the note duration while consonants keep near-natural length, and a closed loop re-measures the output  $f_0$  and corrects residual errors of 0.3–6 st.
4. **Stacking and vocoding.** Five such voices at offsets  $\{0, +12, +7, -12, +3\}$  semitones (octave-folded to stay within  $\pm 15 \text{ st}$  of the source register), each with  $\pm 8$ -cent detune,  $\pm 12 \text{ ms}$  onset jitter and a 5 Hz vibrato ramping in after 150 ms, are summed; the stacked group receives *one* 20-band channel-vocoder pass onto a soft-saw carrier stack playing the same melody notes, and the output is mixed 0.6 vocoded + 0.4 dry. A wordless formant-synthesis choir (pulse-train glottal source through three formant bandpasses) sits underneath as a pad and opens/closes track 10.

On track 10 the melody *is* the loss curve: windows of the validation NLL over epochs  $[e_0, e_1)$  are resampled to 5–7 notes and the normalized loss is mapped onto F harmonic minor — epoch-0 chaos at degree 9, convergence on the tonic — so the choir literally sings gradient descent down to F. New best-validation epochs fire impacts (only the 8 largest improvements), the learning-rate decays at epochs 30 and 60 are the section transitions, and the train/validation gap detunes the pads.

## 6.6 Joint transformer sampler: scorecard

A decoder-only transformer (4 layers,  $d_{\text{model}} = 64$ , 4 heads, context 256) extends the univariate model of §3.3 to the *joint* law of all eight tickers: per-ticker Student- $t$  heads with cross-sectional dependence injected by bootstrapping in-sample rows of the 8-dimensional standardized-residual matrix. Trained on the 3558 common trading days (2012-05-21 to 2026-07-16, post META IPO); best validation NLL 7.4704 (summed over 8 tickers); sample length 4032 days.

The verdict is honest: the transformer matches the joint framework but under-reproduces tails, volatility clustering and (crucially) crisis correlation relative to both the real data and the univariate LSTM — which is why the album’s closer sings the LSTM’s loss curve, not the transformer’s.

## 7 Video synchronization

The video frame at time  $s \in [0, K\tau]$  displays days  $1, \dots, \lfloor sK\delta/(K\tau) \rfloor = \lfloor s\delta/\tau \rfloor$ : the map from audio time to market time is linear with slope  $\delta/\tau$  days per second ( $\approx 63$  trading days/s at

| ticker | ann. vol |       | kurtosis |     | ACF <sub> r </sub> (1) |       | ACF <sub> r </sub> (21) |       | GPH $\hat{d}$ |       |
|--------|----------|-------|----------|-----|------------------------|-------|-------------------------|-------|---------------|-------|
|        | real     | TF    | real     | TF  | real                   | TF    | real                    | TF    | real          | TF    |
| NVDA   | 44.7%    | 37.7% | 10.7     | 5.2 | 0.190                  | 0.045 | 0.092                   | 0.041 | 0.371         | 0.271 |
| AAPL   | 28.3%    | 20.1% | 9.6      | 5.0 | 0.206                  | 0.111 | 0.073                   | 0.038 | 0.395         | 0.334 |
| MSFT   | 26.3%    | 19.8% | 11.3     | 4.4 | 0.232                  | 0.044 | 0.068                   | 0.025 | 0.354         | 0.255 |
| GOOGL  | 27.6%    | 25.2% | 9.6      | 7.3 | 0.138                  | 0.009 | 0.050                   | 0.060 | 0.283         | 0.310 |
| AMZN   | 32.1%    | 25.5% | 8.9      | 5.1 | 0.174                  | 0.077 | 0.063                   | 0.067 | 0.354         | 0.349 |
| META   | 39.8%    | 29.2% | 23.8     | 7.2 | 0.110                  | 0.046 | 0.085                   | 0.044 | 0.305         | 0.355 |
| NFLX   | 45.8%    | 27.9% | 32.4     | 6.8 | 0.129                  | 0.068 | 0.034                   | 0.044 | 0.223         | 0.324 |
| TSLA   | 56.8%    | 44.0% | 7.5      | 5.9 | 0.126                  | 0.054 | 0.060                   | 0.056 | 0.461         | 0.304 |

Table 2: Transformer scorecard, per ticker: real series vs transformer (TF) sample. Real columns use the 8-ticker common sample (post META IPO).

| measure                                 | real  | transformer sample |
|-----------------------------------------|-------|--------------------|
| full-sample corr(NVDA, AAPL)            | 0.484 | 0.266              |
| crisis corr (top-decile joint-vol days) | 0.640 | 0.213              |

  

| metric (NVDA)           | real (docs) | LSTM  | transformer |
|-------------------------|-------------|-------|-------------|
| annualized vol          | 45.2%       | 53.4% | 37.7%       |
| kurtosis                | 9.8         | 10.1  | 5.2         |
| ACF <sub> r </sub> (1)  | 0.171       | 0.131 | 0.045       |
| ACF <sub> r </sub> (21) | 0.092       | 0.089 | 0.041       |
| GPH $\hat{d}$           | 0.397       | 0.516 | 0.271       |

Table 3: Left: cross-sectional dependence, real vs transformer sample. Right: NVDA baseline comparison against the univariate LSTM of §3.3 (the real column there uses the full 2010–2026 sample, so values differ slightly from Table 2).

$B = 128$ ,  $\delta = 3$ ). Overlays are the regime shading  $\{\beta_t = 1\}$  (visible iff the playhead has passed the region’s onset), the running price  $e^{X_t}$  on a log axis,  $v_t$  in a lower panel, and markers at  $\{t : |r_t| > 4\hat{s}\}$  — the visual twins of the breakdown, the filter curve, and the crash cymbal respectively.

## Caveat

A flat return spectrum (§2.2) is exactly the statement that the composition carries no linear predictive content. Apollo is prophecy only in the sense that mattered at Delphi: accurate, resonant, and of no use for trading.